



BAULKHAM HILLS HIGH SCHOOL

2022

HSC Assessment Task 1

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using non-erasable black or blue pen.
- Calculators approved by NESA may be used.
- Marks may be deducted for careless or badly arranged work.
- Relevant formulae from the NESA Reference Sheet are provided on page 2.
- Answer all questions on the appropriate page of your answer booklet.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.
- If extra paper is used, it must be clearly labelled with your NESA number and the question being answered, and stapled inside your answer booklet.

Total marks – 47

Exam consists of 7 pages.

Section I: Pages 2 - 4

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Section II: Pages 5 - 7

42 marks

Attempt Questions 6 – 8

Allow about 52 minutes for this section

Complex Numbers

$$z = a + ib = r(\cos \theta + i \sin \theta) \\ = re^{i\theta}$$

$$\left[r(\cos \theta + i \sin \theta) \right]^n = r^n(\cos n\theta + i \sin n\theta) \\ = r^n e^{in\theta}$$

Section I:

Marks

5 marks

Attempt Questions 1 - 5

Use the multiple choice answer sheet in your booklet.

1. If $w = 5 - 2i$, which of the following is $\frac{1}{2-w}$? 1

(A) $\frac{3}{13} - \frac{2}{13}i$

(B) $-\frac{3}{13} + \frac{2}{13}i$

(C) $-\frac{3}{13} - \frac{2}{13}i$

(D) $\frac{3}{13} + \frac{2}{13}i$

2. For the complex numbers z and w , where $\arg(z) = \frac{\pi}{2}$ and $\arg(w) = \frac{\pi}{4}$, 1

the value of $\arg\left(\frac{z^5}{w^4}\right)$ is:

(A) $-\frac{\pi}{2}$

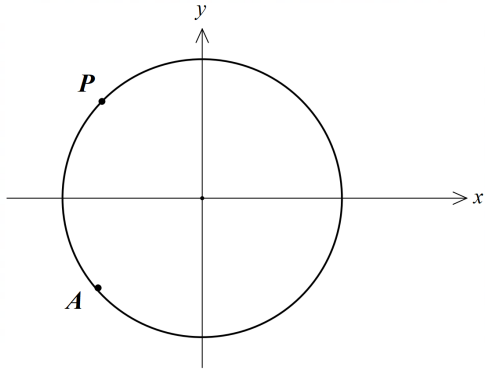
(B) $\frac{\pi}{2}$

(C) π

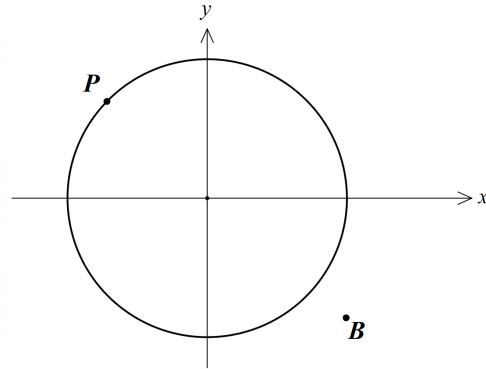
(D) $\frac{\pi}{4}$

3. If $\overrightarrow{OP} = z$, $\frac{\pi}{2} < \arg z < \pi$ and $|z| < 1$ then which of the following is a possible location for the complex number $\sqrt{z}$?

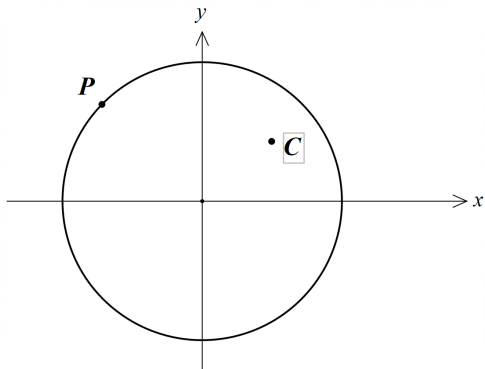
(A) $\overrightarrow{OA}$



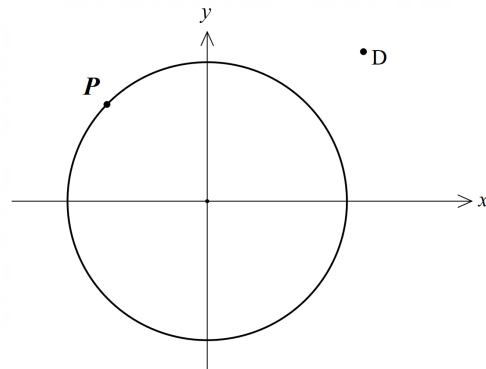
(B) $\overrightarrow{OB}$



(C) $\overrightarrow{OC}$



(D) $\overrightarrow{OD}$



4. The polynomial equation $P(z) = 0$ has real coefficients, and has roots which include $z = -2 + i$ and $z = 2$. The minimum degree of $P(z)$ would be:

- (A) 1
(B) 2
(C) 3
(D) 4

5. The points corresponding to the four complex numbers given by

1

$$z_1 = 2\text{cis}\left(\frac{\pi}{3}\right), \quad z_2 = \text{cis}\left(\frac{3\pi}{4}\right), \quad z_3 = 2\text{cis}\left(\frac{-2\pi}{3}\right), \quad z_4 = \text{cis}\left(\frac{-\pi}{4}\right)$$

are the vertices of a parallelogram in the complex plane. Which one of the following statements is **not** true?

- (A) The acute angle between the diagonals of the parallelogram is $\frac{5\pi}{12}$
- (B) The diagonals of the parallelogram have lengths 2 and 4
- (C) $z_1 z_2 z_3 z_4 = 0$
- (D) $z_1 + z_2 + z_3 + z_4 = 0$

End of Section I

Section II: 42 marks

Attempt Questions 6 - 8

Answer each question on the appropriate page of the answer booklet, showing all necessary working

Question 6 (14 marks)

a) Let $z = a + ib$, where a and b are real, find:

(i) $\operatorname{Im}(4i - z)$ 1

(ii) $\overline{(3iz)}$ in the form of $x + iy$, where x and y are real numbers 1

b) If $wz = 3 - i$, where $w = 1 - i$ find:

(i) z in the form of $x + iy$, where x and y are real numbers 2

(ii) $\arg w$ 1

c) (i) Express $3 - \sqrt{3}i$ in polar form. 2

(ii) Express $(3 - \sqrt{3}i)^3$ in the form of $x + iy$, where x and y are real numbers. 1

(iii) Find the integer values of n for which $(3 - \sqrt{3}i)^n$ is real. 2

d) Sketch the locus of z on separate complex planes:

(i) $|z| = |z + 1 - \sqrt{3}i|$ 2

(ii) $\frac{-2\pi}{3} < \arg(z + 1) \leq 0$ 2

Question 7 (14 marks)

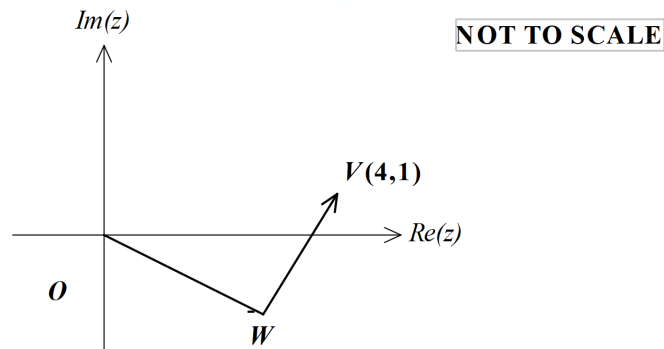
- a) Given that $(2 + 3i)$ is one of the roots of the polynomial $P(x) = 4x^3 - 13x^2 + 40x + 39$, 3
solve $4x^3 - 13x^2 + 40x + 39 = 0$ over the complex field.
- b) (i) Find the square roots of $-8 - 6i$, giving the answers in the form of $x + iy$, where x 2
and y are real numbers.
- (ii) Hence, or otherwise, solve the equation $2z^2 - 8z + 12 + 3i = 0$, giving your solutions 2
in the form $a + ib$ where a and b are real numbers.
- c) (i) If ω is a non-real cube root of unity, show that $1 + \omega + \omega^2 = 0$. 1
- (ii) Hence prove that $\frac{1}{1 + \omega} - \frac{1}{1 + \omega^2} = -(1 + 2\omega)$. 2
- d) (i) Express $z = -1 - \sqrt{3}i$ in the form of $re^{i\theta}$. 2
- (ii) Hence, find the real values of a and b such that $-1 - \sqrt{3}i = e^{a+ib}$. 2

Question 8 (14 marks)

- a) (i) Sketch on an Argand diagram, the locus of the point P which satisfies the equation $|z - 3| = 3$. 2

- (ii) Use the diagram to show that P satisfies the condition $\arg(z - 3) = \arg z^2$. 2

- b) The point V represents the complex number $4 + i$, $\angle OWV = \frac{\pi}{2}$ and $|VW| = |OW|$. Find the complex number represented by the point W in the form of $a + ib$ where a and b are real numbers. 2



- c) (i) By using De Moivre's theorem and the binomial expansion of $(\cos \theta + i \sin \theta)^4$, show that $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$. 3
- (ii) Hence solve the equation $8x^4 - 8x^2 + 1 = 0$. 2
- (iii) Hence find the exact value of $\cos \frac{\pi}{8}$. 3

END OF EXAMINATION

Extension-2 Task-1, 2022 Solutions.

Section-1

Multiple Choice:

$$1) \frac{1}{2-w} = \frac{1}{2-5+2i} = \frac{1}{-3+2i} \times \frac{-3-2i}{-3-2i} = \frac{-3-2i}{(-3)^2-(2i)^2} = \frac{-3}{13} - \frac{2}{13}i \quad \boxed{C}$$

$$\begin{aligned} 2) \arg\left(\frac{z^5}{w^4}\right) &= \arg z^5 - \arg w^4 \\ &= 5\arg z - 4\arg w \\ &= 5 \times \frac{\pi}{2} - 4 \times \frac{\pi}{4} = \frac{3\pi}{2} = -\frac{\pi}{2} \end{aligned} \quad \boxed{A}$$

$$3) \text{ Since } |z| < 1, \text{ therefore } |\sqrt{z}| > 1$$

so $\sqrt{z}$ lies outside the circle.

$$\text{Also if } \frac{\pi}{2} < \arg z < \pi \text{ then } 0 < \arg \sqrt{z} < \frac{\pi}{2} \quad \boxed{D}$$

$$\text{as } \arg(\sqrt{z}) = \frac{1}{2}\arg z$$

$$4) \text{ Since } P(z) \text{ has real coeff. } \therefore -2+i, -2-i \text{ and } z \text{ are roots.}$$

$$\text{Therefore minimum degree of } P(z) \text{ is } 3. \quad \boxed{C}$$

$$5) |z_1| = 2, \arg z_1 = \frac{\pi}{3}$$

$$|z_3| = 2, \arg z_3 = -\frac{2\pi}{3}$$

$$|z_2| = 1, \arg z_2 = \frac{3\pi}{4}$$

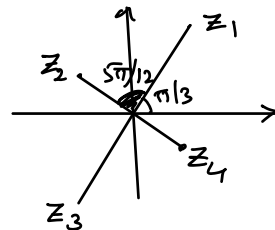
$$|z_4| = 1, \arg z_4 = -\frac{\pi}{4}$$

A Correct

B Correct

C Incorrect as $z_1 z_2 z_3 z_4 \neq 0$

D Correct



$\boxed{C}$

Section II

Question- 6

a) i) $z = a + ib$

(1 mark correct answer.)

$$4i - z = 4i - a - ib$$

$$= -a + (4-b)i$$

$$\text{Im}(4i - z) = 4 - b$$

ii) $\overline{3iz} = \overline{3i(a+ib)}$

(1 mark correct answer.)

$$= \overline{i3a - 3b}$$

$$= -3b - 3ia$$

b) i) $(1-i)z = 3-i$

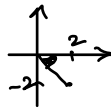
$$z = \frac{3-i}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{4+2i}{2}$$

$$= 2+i$$

{ 2 marks correct answer.
1 mark to make z as subject
and multiply & divide by $(1-i)$

ii) $\arg z = -\frac{\pi}{4}$

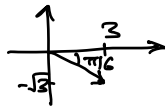


(1 mark correct answer)

c) i) Let $z = 3 - \sqrt{3}i$

$$|z| = \sqrt{(3)^2 + (-\sqrt{3})^2} = \sqrt{12} = 2\sqrt{3}$$

$$\arg z = -\frac{\pi}{6}$$



{ 2 marks correct solution
1 mark either correct
 $|z|$ or $\arg z$

$$\therefore z = 2\sqrt{3} \text{ Cis}(-\frac{\pi}{6})$$

ii) $(3 - \sqrt{3}i)^3 = [2\sqrt{3} \text{ Cis}(-\frac{\pi}{6})]^3$

{ 1 mark correct solution.

$$= 24\sqrt{3} \text{ Cis}(-\frac{\pi}{2})$$

$$= -24\sqrt{3}i$$

Question 6 (continues)

c) iii) $(3 - \sqrt{3}i)^n = (2\sqrt{3})^n \text{cis}(-\frac{n\pi}{6})$

$$= (2\sqrt{3})^n \left(\cos \frac{n\pi}{6} - i \sin \frac{n\pi}{6} \right)$$

For real; $\sin \frac{n\pi}{6} = 0$

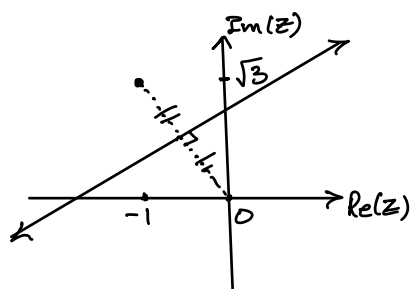
$$\frac{n\pi}{6} = k\pi, \text{ where } k \text{ is integer.}$$

$$\therefore n = 6k$$

$\therefore n$ is any integral multiple of 6.

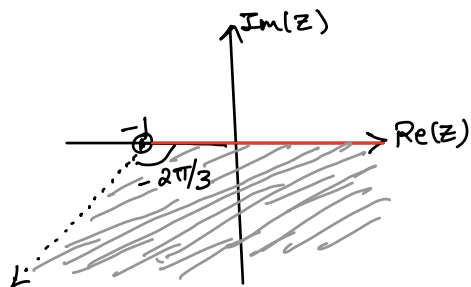
2 marks
correct solution
1 mark attempt
to equate imaginary
part equal to zero.

d) i)



2 marks correct locus with
all important features.
1 mark to recognise it is a
perp. bisector of two incorrect
points or finding the algebraic
equation for the perp. bisector
line or either correct bisector/perp.
of two given points.

ii)



2 marks correct locus with
all important features
1 mark for correct boundaries
or shade the correct region

Question-7

- a) Since $2+3i$ is a root and all the coefficients of poly. $P(x)$ are real, therefore its conjugate $(2-3i)$ must also be a root.

Let α be the third root of $P(x) = 4x^3 - 13x^2 + 40x + 39$

$$\text{Sum of roots} = -\frac{b}{a} = \frac{13}{4}$$

$$2+3i + 2-3i + \alpha = \frac{13}{4}$$

$$\alpha = \frac{13}{4} - 4$$

$$\alpha = -\frac{3}{4}$$

Hence three roots are; $2+3i$, $2-3i$, $-\frac{3}{4}$

3 marks correct solution
2 marks use correct method to attempt to solve poly $P(x)=0$
1 mark to identify $2-3i$ as another root.

- b) i) Let $z = x + iy$, where $x, y \in \mathbb{R}$

$$\therefore (x+iy)^2 = -8-6i$$

$$x^2 - y^2 + 2ixy = -8 - 6i$$

Equate real and imaginary coefficients:

$$x^2 - y^2 = -8 \quad \text{--- (1)} \quad \& \quad 2xy = -6$$

$$xy = -3 \quad \text{--- (2)}$$

Solve (1) & (2) simultaneously:

$$x^2 - \left(-\frac{3}{x}\right)^2 = -8$$

$$x^2 - \frac{9}{x^2} = -8$$

$$x^4 + 8x^2 - 9 = 0$$

$$(x^2+9)(x^2-1) = 0$$

$$x^2 - 1 = 0 \quad (\because x^2 \neq -9 \text{ as } x \in \mathbb{R})$$

$$x = \pm 1$$

$$y = \mp 3$$

$$\therefore z = \pm(1-3i)$$

2 marks correct solution
1 mark to obtain two simultaneous equations and attempt to solve or any other equivalent method.

Question 7 (continues)

b) ii) $2z^2 - 8z + 12 + 3i = 0$

$$\Delta = b^2 - 4ac$$

$$= (-8)^2 - 4(2)(12 + 3i)$$

$$= 64 - 96 - 24i$$

$$= -32 - 24i$$

$$= 4(-8 - 6i)$$

$$\therefore z = \frac{8 \pm 2(1 - 3i)}{4}$$

$$= \frac{8 + 2 - 6i}{4}, \frac{8 - 2 + 6i}{4}$$

$$= \frac{5}{2} - \frac{3i}{2}, \frac{3}{2} + \frac{3i}{2}$$

{ 2 marks correct solution
1 mark to find discriminant
or attempt to link results
from part i) using quad.
formula.

c) i) Let z is cube root of 1

then $z^3 = 1$

$$z^3 - 1 = 0$$

$$(z - 1)(z^2 + z + 1) = 0$$

Since w is non real root

$$\therefore w^2 + w + 1 = 0 \quad (\because w \neq 1)$$

(1 mark correct solution)

ii) $\frac{1}{1+w} - \frac{1}{1+w^2} = \frac{1}{-w^2} - \frac{1}{-w}$

($\because 1+w = -w^2$ and $1+w^2 = -w$)

$$= -\frac{1}{w^2} + \frac{1}{w}$$

$$= -\frac{1+w}{w^2} \times \frac{w}{w}$$

$$= -w + w^2$$

$$= -w - w - 1$$

$$= -2w - 1$$

Hence proved.

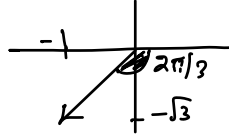
{ 2 marks correct proof
1 mark attempt to
prove by using part i)

Question 7 (Continues)

d) i) $z = -1 - \sqrt{3}i$

$$|z| = \sqrt{(-1)^2 + (-\sqrt{3})^2} = 2$$

$$\arg z = -\frac{2\pi}{3}$$



$$\therefore z = 2e^{-i2\pi/3}$$

2 marks correct solution
1 mark either correct $|z|$ or $\arg z$

ii) $-1 - \sqrt{3}i = e^{a+ib}$

$$2e^{-i2\pi/3} = e^a \cdot e^{ib}$$

Equating both sides;

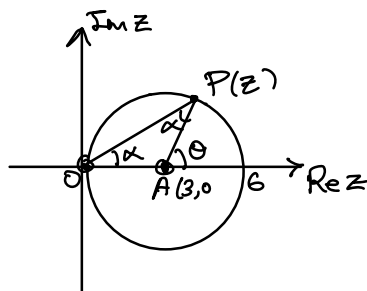
$$2 = e^a \quad \text{and} \quad b = -\frac{2\pi}{3}$$

$$a = \ln 2, \quad b = -\frac{2\pi}{3}$$

2 marks correct answer
1 mark either correct a or b

Question - 8

a) i)



2 marks correct locus
1 mark either correct centre or radius of circle

ii) $\arg(z-3) = 0$

$$\arg z = \alpha$$

$$\angle OPA = \alpha \quad (\text{OA} = \text{AP}, \text{ opp. angles to equal sides of isosceles } \triangle)$$

$$\therefore \theta = 2\alpha \quad (\text{Ext angle of } \triangle OPA \text{ is equal to sum of int. opp. } \angle\text{'s})$$

$$\therefore \arg(z-3) = 2 \arg z \quad \text{or} \quad \arg(z-3) = \arg z^2$$

2 marks correct proof
1 mark attempt to prove by using geometry

Question 8 (continues)

b) let $W = x + iy$

$$\vec{OW} = -i \vec{WV}$$

$$\begin{aligned} x + iy &= -i(4 + i - x - iy) \\ &= -4i + 1 + ix - y \\ &= (1 - y) + i(-4 + x) \end{aligned}$$

Equating real & imaginary coefficients:

$$x = 1 - y \quad \text{--- ①} \quad y = -4 + x \quad \text{--- ②}$$

$$y = -4 + 1 - y$$

$$y = -\frac{3}{2}, \quad x = \frac{5}{2}$$

$$\therefore W = \frac{5}{2} - i\frac{3}{2}$$

2 marks correct answer
1 mark attempt to relate $\vec{OW}$ and $\vec{WV}$ and solve the equation.

c) i) By De Moivre theorem:

$$(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$$

By Binomial expansion;

$$(\cos \theta + i \sin \theta)^4 = c^4 + 4c^3(is) + 6c^2(i^2s^2) + 4c(is)^3 + (is)^4$$

$$\text{where } c = \cos \theta, \quad s = \sin \theta$$

$$= (c^4 - 6c^2s^2 + s^4) + i(4c^3s - 4cs^3)$$

Equating real parts;

$$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$= \cos^4 \theta - 6\cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= \cos^4 \theta - 6\cos^2 \theta + 6\cos^4 \theta + 1 - 2\cos^2 \theta + \cos^4 \theta$$

$$= 8\cos^4 \theta - 8\cos^2 \theta + 1$$

3 marks correct solution
2 marks for correct expansion and equates real part
1 mark for both correct expansion

Question-8 (Continues)

c) ii) To solve $8x^4 - 8x^2 + 1 = 0$

$$\text{Let } x = \cos \theta; \therefore \cos 4\theta = 0$$

$$4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$$

$$\therefore x = \cos \frac{\pi}{8}, \cos \frac{3\pi}{8}, \cos \frac{5\pi}{8}, \cos \frac{7\pi}{8}$$

2 marks correct solution
1 mark attempt to solve the equation by letting $x = \cos \theta$ and attempting to solve $\cos 4\theta = 0$

iii) $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$ are solutions of $8\cos^4\theta - 8\cos^2\theta + 1 = 0$

Also, by quad. formula

$$\cos^2\theta = \frac{8 \pm \sqrt{(-8)^2 - 4 \times 8 \times 1}}{16}$$

$$= \frac{8 \pm \sqrt{32}}{16}$$

$$= \frac{2 \pm \sqrt{2}}{4}$$

$$\cos \theta = \pm \frac{\sqrt{2 \pm \sqrt{2}}}{2}$$

Since $\frac{\pi}{8}$ is in Quadrant I $\therefore \cos \frac{\pi}{8}$ will be positive

and either $\frac{\sqrt{2+\sqrt{2}}}{2}$ or $\frac{\sqrt{2-\sqrt{2}}}{2}$

$$\text{As } \cos \frac{\pi}{8} > \cos \frac{3\pi}{8} \quad \text{Hence } \cos \frac{\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$$

3 marks correct solution
2 marks uses the fact $\cos \theta > 0$ OR $\cos \frac{\pi}{8} > \cos \frac{3\pi}{8}$ to identify exact value OR provides exact value with no justification
1 mark uses quad. formula to find the exact value of $\cos^2\theta$.

END OF SOLUTIONS